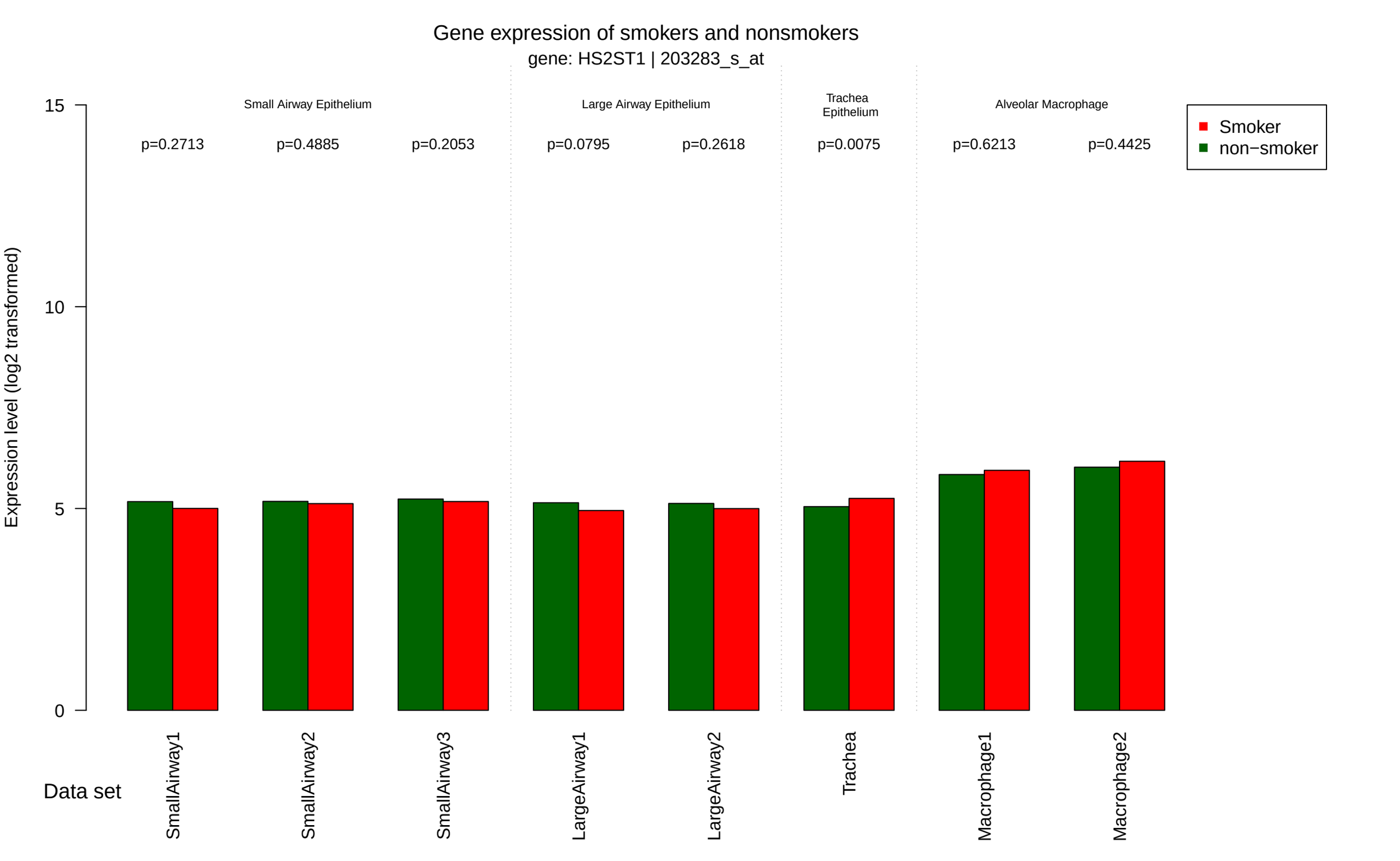
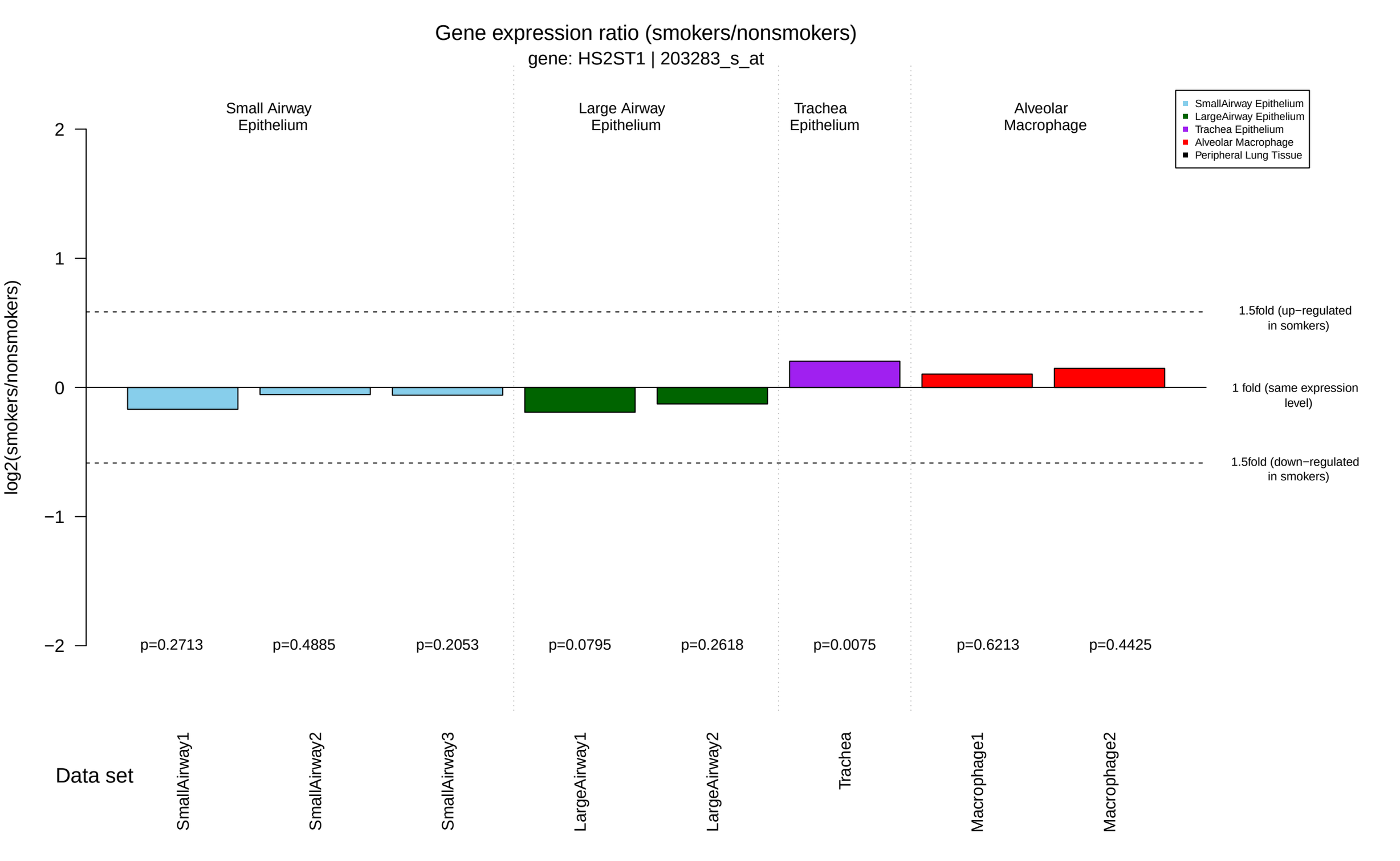
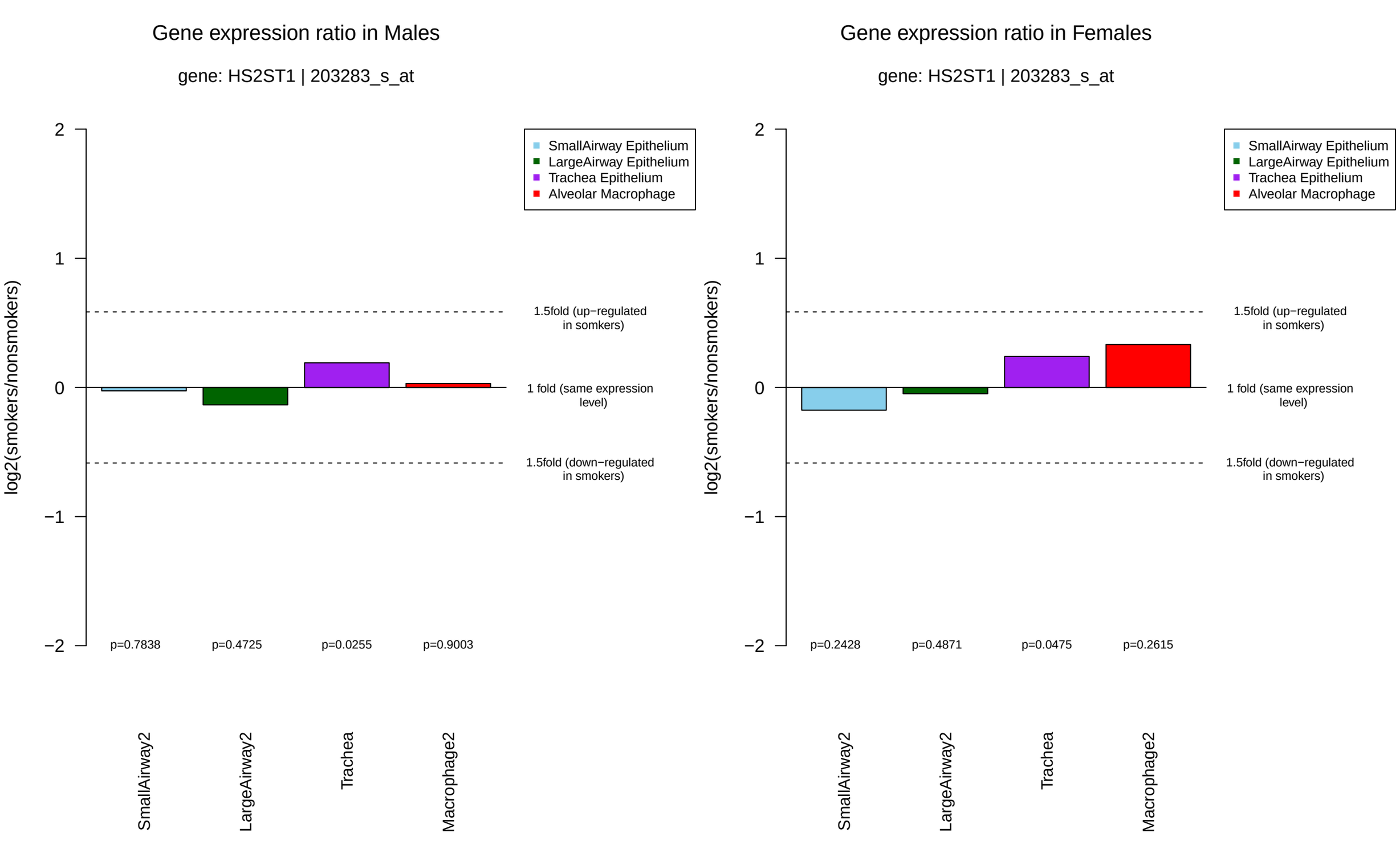
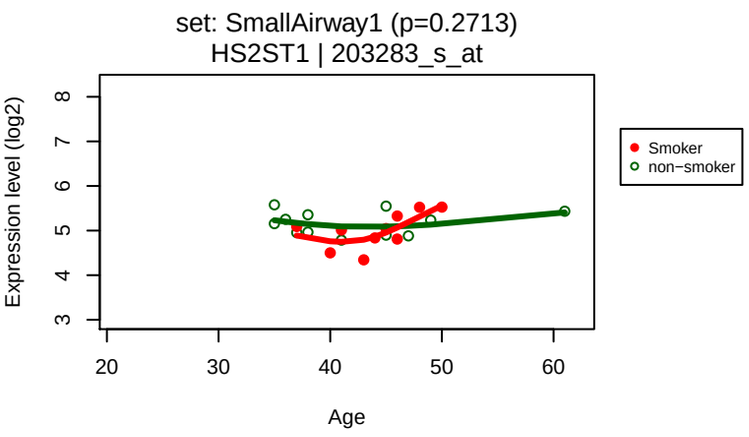
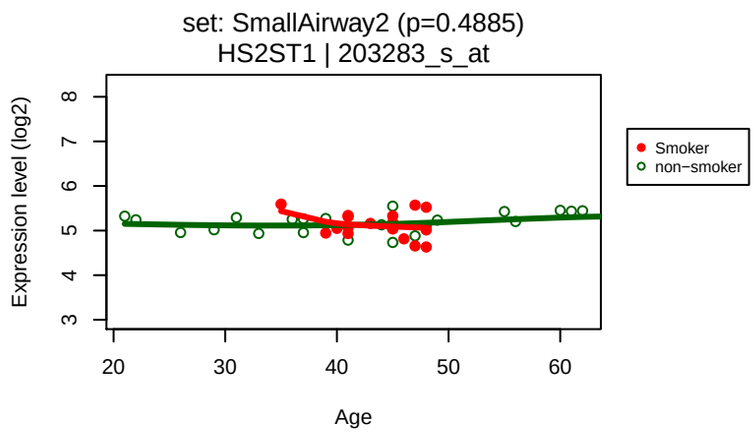
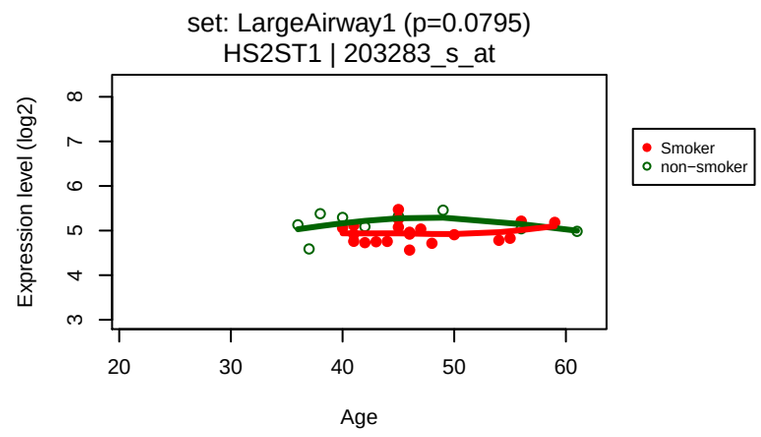

Gene expression of smokers and nonsmokers
gene: HS2ST1 | 203283_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2713
p=0.4885
p=0.2053
p=0.0795
p=0.2618
p=0.0075
p=0.6213
p=0.4425
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

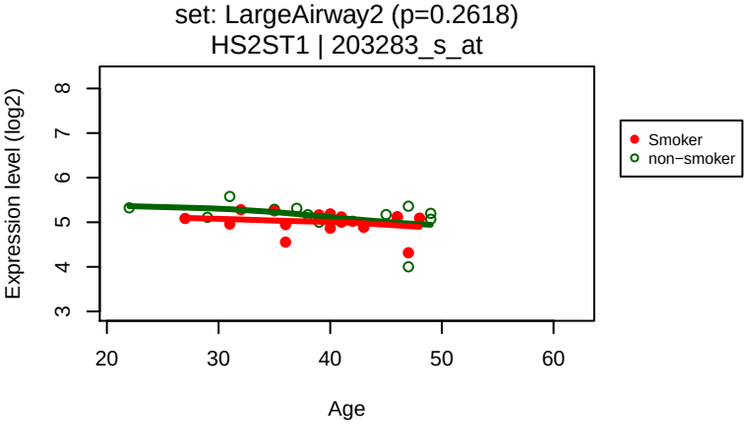

Gene expression ratio (smokers/nonsmokers)
gene: HS2ST1 | 203283_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2713
p=0.4885
p=0.2053
p=0.0795
p=0.2618
p=0.0075
p=0.6213
p=0.4425
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

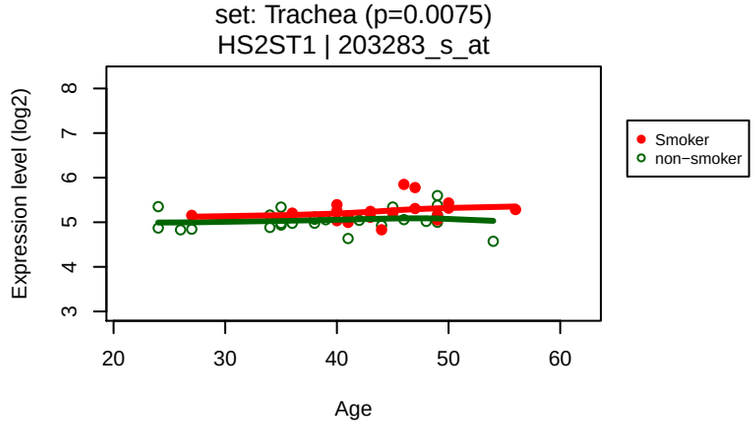

Gene expression ratio in Males
gene: HS2ST1 | 203283_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7838
p=0.4725
p=0.0255
p=0.9003
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HS2ST1 | 203283_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2428
p=0.4871
p=0.0475
p=0.2615
SmallAirway2
LargeAirway2
Trachea
Macrophage2

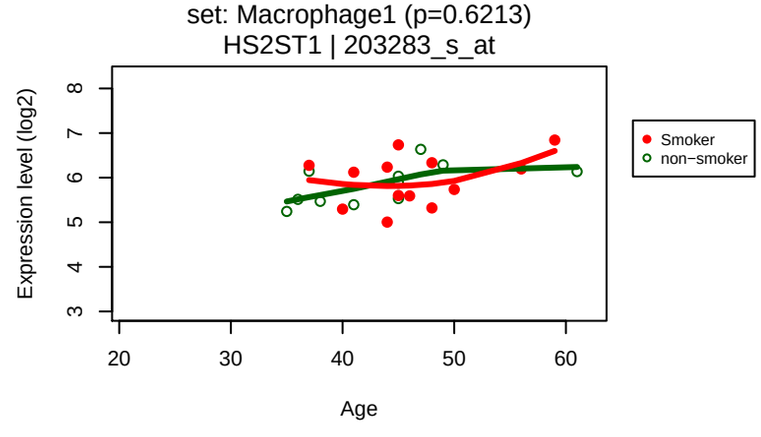

set: SmallAirway1 (p=0.2713)
HS2ST1 | 203283_s_at
Expression level (log2)
Age
Smoker
non−smoker

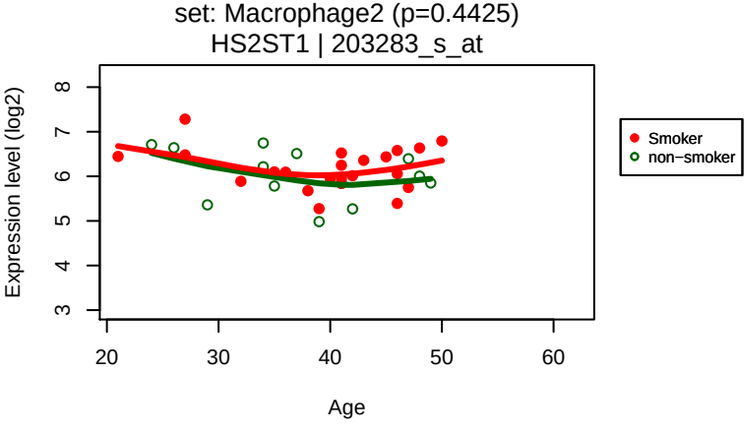

set: SmallAirway2 (p=0.4885)
HS2ST1 | 203283_s_at
Expression level (log2)
Age
Smoker
non−smoker

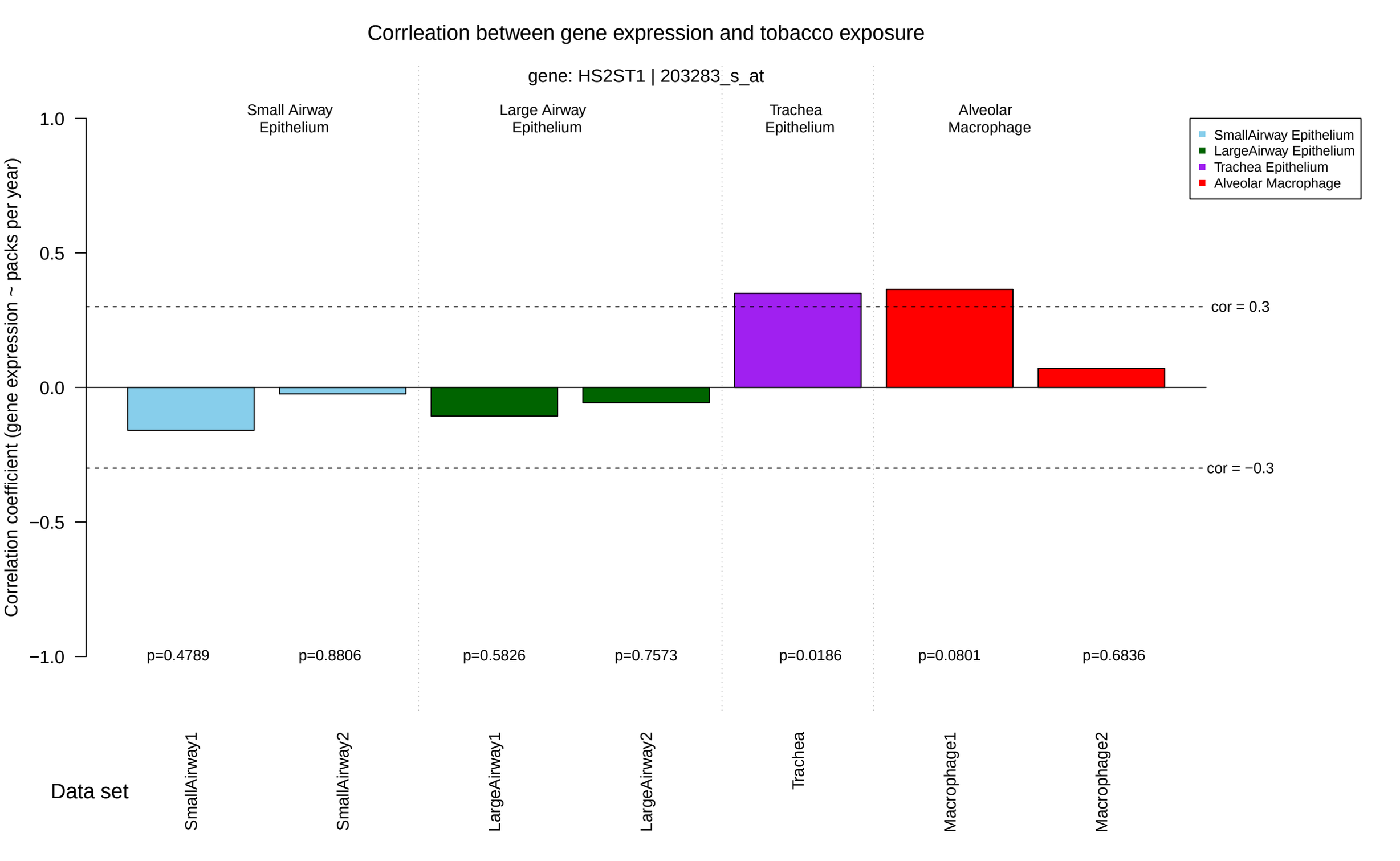

set: LargeAirway1 (p=0.0795)
HS2ST1 | 203283_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2618)
HS2ST1 | 203283_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0075)
HS2ST1 | 203283_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6213)
HS2ST1 | 203283_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4425)
HS2ST1 | 203283_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HS2ST1 | 203283_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4789
p=0.8806
p=0.5826
p=0.7573
p=0.0186
p=0.0801
p=0.6836
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2